Polynomials

The big idea

Polynomials are algebraic expressions like $3x^2 - 2x + 7$ with whole-number powers. This chapter learns to classify them, find their **zeroes**, use the **Remainder** and **Factor** theorems, and factorise using **identities**.

Vocabulary

- **Degree**: highest power. Linear (1), quadratic (2), cubic (3). A non-zero constant has degree 0; the zero polynomial has no degree.
- By number of terms: **monomial** (1), **binomial** (2), **trinomial** (3).
- $x + 1/x$, $\sqrt{x} + 3$ are **not** polynomials (powers -1 and $\frac{1}{2}$).

Zeroes

A **zero** of $p(x)$ is a value c with $p(c) = 0$. A linear polynomial $ax + b$ has exactly one zero, $-b/a$. A polynomial can have more than one zero; the zero polynomial has every number as a zero.

The Remainder Theorem

REMAINDER THEOREM

If $p(x)$ is divided by $(x - a)$, the remainder is **$p(a)$**.

FIND THE REMAINDER WHEN $X^3 + 3X^2 + 3X + 1$ IS DIVIDED BY $X + 1$

$x + 1 = x - (-1)$, so the remainder is $p(-1) = -1 + 3 - 3 + 1 = \mathbf{0}$.

The Factor Theorem

FACTOR THEOREM

$(x - a)$ is a factor of $p(x)$ **if and only if** $p(a) = 0$.

FACTORISE $X^3 - 23X^2 + 142X - 120$

Try factors of 120: $p(1) = 1 - 23 + 142 - 120 = 0$, so $(x - 1)$ is a factor.

Divide: $x^3 - 23x^2 + 142x - 120 = (x - 1)(x^2 - 22x + 120) =$ **$(x - 1)(x - 10)(x - 12)$**.

To factorise $ax^2 + bx + c$, split the middle term: find p and q with $p + q = b$ and $pq = ac$.

Algebraic identities

IDENTITIES TO KNOW

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x - y)^2 = x^2 - 2xy + y^2$$

$$x^2 - y^2 = (x + y)(x - y)$$

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$$

$$(x + y)^3 = x^3 + y^3 + 3xy(x + y);\ (x - y)^3 = x^3 - y^3 - 3xy(x - y)$$

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

SPECIAL CASE

If $x + y + z = 0$, then **$x^3 + y^3 + z^3 = 3xyz$**. So $(-12)^3 + 7^3 + 5^3 = 3(-12)(7)(5) = -1260$.

EVALUATE 104×96 WITHOUT MULTIPLYING DIRECTLY

$(100 + 4)(100 - 4) = 10000 - 16 =$ **9984**.

Quick recap

- Degree, zeroes, types of polynomials.
- Remainder theorem: remainder = p(a). Factor theorem: p(a) = 0 $\Leftrightarrow$ (x – a) is a factor.
- Identities make factorising and calculating fast.

Practice questions

1. The degree of $5x^3 + 4x^2 + 7x$ is
   (a) 1 (b) 2 (c) 3 (d) 5

2. Which is a polynomial?
   (a) $x + 1/x$ (b) $\sqrt{x} + 2$ (c) $3x^2 - 4x + 1$ (d) $1/(x + 1)$

3. The remainder when $x^2 + 2x + 1$ is divided by $x - 1$ is
   (a) 0 (b) 2 (c) 4 (d) 1

4. If $(x - 2)$ is a factor of $x^2 + kx - 6$, then k equals
   (a) 2 (b) 1 (c) -1 (d) 3

5. If $a + b + c = 0$, then $a^3 + b^3 + c^3$ equals
   (a) 0 (b) $3abc$ (c) abc (d) $a + b + c$

Answers: 1. c 2. c 3. c 4. b 5. b